SAVITRIBAI PHULE PUNE UNIVERSITY

A PROJECT REPORT ON

Decoding Public Opinion: Sentiment Classification of Hinglish and Emoji-Enriched Reddit Discussions on Government Policies

SUBMITTED TOWARDS THE
PARTIAL FULFILLMENT OF THE REQUIREMENTS OF

BACHELOR OF ENGINEERING (Computer Engineering)

BY

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CERTIFICATE

This is to certify that the Project Entitled

Decoding Public Opinion: Sentiment Classification of Hinglish and Emoji-Enriched Reddit Discussions on Government Policies

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Abstract

Websites like Reddit have become popular for political conversations, debates, and opinion formations. One interesting thing is that Indian users frequently participate in their own unique code-mixed style that deploys simple Hinglish, and is overlain with emojis that add emotional extra layers and make the linguistic construction even more complex. It is precisely these informal, multilingual, and emoji-embedded chats that traditional training algorithms, often focused on singular languages and formality, tend to lose track of.

This work proposes detecting and classifying sentiment in Hinglish and emoji-saturated discussions pertaining to different policies of the Indian government including, but not limited to, taxation, infrastructure reforms, digital governance, welfare schemes, and national security. Collections of Reddit discussions form the core datasets. These datasets go through a great deal of curation with respect to code-mixing, emojis, transliterations, informal syntax, and the assorted complexities of informal articulation.

We customize a number of transformers including mBERT, XLM-R, MuRIL, HingBERT, and the FNet-upgraded hybrid models geared toward multilingual sentiment. Emojis were included, and merged through layer formation, effectively as informative indicators.

Indian language-based models outperform findings from studies in multilingual models.

The designed system porvides a live stream inter tface in which users can predict the sentiment –positive and negative/ neutral– from Hinglish political discourse.

It enhances the understanding of the public reaction towards governing policy

Keywords: Hinglish. Code-mixed text. Emojis. Public opinion mining. Goverment policy sentiment. Reddit analysis. Transforemrs. mBERT. MuRIL. FNET.

Acknowledgments

It gives us great pleasure in presenting the preliminary project report on ***'Decoding Public Opinion: Sentiment Classification of Hinglish and Emoji-Enriched Reddit Discussions on Government Policies'****.*

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Chapter 1

Synopsis

1.1 Project Title

Decoding Public Opinion: Sentiment Classification of Hinglish and Emoji-Enriched Reddit Discussions on Government Policies

1.2 Project Option

Type: Internal Project

This project is carried out within the institute as part of the academic curriculum.

1.3 Internal Guide

Prof. Chitraleka Dwivedi

1.4 Sponsorship and External Guide

Sponsorship: No external sponsorship.

1.5 Technical Keywords (As per ACM Keywords)

1. I. Computing Methodologies

(a) Natural Language Processing (NLP)
(b) Deep Learning
(c) Transformer Models
(d) Emotion Recognition
(e) Sentiment Analysis
(f) Representation Learning

II. Information Systems

(a) Social Media Analytics
(b) Opinion Mining
(c) Data Mining

III. Related Concepts

1. Emoji Semantics
2. Code-Mixed Language Processing
3. Hinglish Text Normalization
4. Multimodal Sentiment

1.6 Problem Statement

Political discussions on platforms like Reddit India are highly multilingual and emotion-rich. Users frequently combine **Hindi + English** with Devanagari insertions (e.g., "Modi decision sahi ") and heavily use emojis to reinforce attitude ("", "", "", "", "").

Traditional sentiment analysis models fail because:

1. They assume **monolingual, grammatically clean text**.
2. They cannot interpret **code-mixing patterns** such as "Nayi policy but implementation weak ".

3. Emojis, which hold strong emotional meaning, are ignored or incorrectly treated as noise.

4. Government policy-related discussions are highly **contextual and political**, requiring domain-specific understanding.

Therefore, there is a need to develop a transformer-based sentiment classifier tailored to:

- Hinglish code-mixed text
- Emoji-enriched expressions
- Reddit political discourse
- Government policy–related opinion mining

Multiple studies confirm transformers outperform older ML models for code-mixed sentiment tasks [1], [3], [7], [9], [12], [15], [17].

1.7 Abstract

This study builds a transformer-based setup to classify sentiment within Hinglish with emojis on Reddit posts about Indian government policy discussions. Code-switching with emojis requires normalization, transliteration, and lexicon augmentation and embedding fusion methods for processing. mBERT, XLM-R, MuRIL, FNet and HingBERT were tested. Results show that India-focused multilingual models surpass general models on Hinglish political opinion mining.

Government policy, Reddit, mBERT, MuRIL, FNET, Transformers.*Keywords: Hinglish, Code-switching, Emojis, Opinion Mining, Sentiment analysis on Government policy, Reddit, mBERT, MuRIL, FNET, Transformers.*

1.8 Goals and Objectives

To decode public opinion on government policies by analyzing Hinglish + emoji enriched Reddit discussions using transformer-based sentiment classification.

Objectives

1. *Collect public opinion discussions from Reddit r/India, r/IndianPolitics, r/IndiaSpeaks, r/Chodi, etc.*
2. *Clean and preprocess Hinglish with mixed scripts (Devanagari + Roman).*
3. *Convert emojis to semantic tags using lexicons and embedding fusion.*
4. *Build a labeled dataset for positive, negative, neutral sentiments.*
5. *Fine-tune multilingual and Indian-language transformer models.*
6. *Evaluate using metrics like Accuracy, F1 Score, MCC, Hamming Loss.*
7. *Deploy a user-friendly interface for real-time sentiment prediction.*

1.9 Relevant mathematics associated with the Project

Let X be input text and y be sentiment label:

$$f(X) = y \in \{Positive, Negative, Neutral\}$$

2. F1-Score

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

3. Emoji Polarity Weighting

$$S_{final} = S_{text} + \alpha \cdot S_{emoji}$$

Where α is the emoji influence coefficient.

4. Cross-Entropy Loss

$$L = -\sum y_i \cdot \log(\hat{y}_i)$$

These are widely used in recent deep learning–based sentiment research [4], [6], [10], [15], [19].

1.10 Names of Conferences / Journals where papers can be published

IEEE

- *IEEE Access*
- *IEEE Transactions on Affective Computing*

ACM

- *ACM TALLIP (Transactions on Asian and Low-Resource Language Information Processing)*

Springer

- *Neural Computing & Applications*
- *Social Network Analysis & Mining*

Elsevier

- *Expert Systems with Applications*

Conferences

- *ACL*
- *EMNLP*
- *LREC*
- *ICON*
- *FIRE*

1.11 Review of Conference/Journal Papers supporting Project idea

1. ***Toward Emotion Recognition in Hindi-English Code-Mixed Data*** *(2021) – Transformer-based Hinglish emotion analysis [1].*
2. ***Transformer-Based Sentiment Analysis on Code-Mixed Data*** *(2022) – Demonstrated transformer superiority [2].*

3. ***Code-Mixed Text Sentiment Analysis Review** (2020) – Challenges of code-mixing documented [3].*
4. ***Emoji-Fused Sentiment (ML + BERT)** (2022) – Modeling emoji emotion intensities [4].*
5. ***Sentiment Analysis of Code-Mixed Social Media Text** (2021) – Language mixing + social media context [5].*
6. ***Hinglish Cross-Lingual Sentiment Analysis** (2020) – Cross-lingual transformer strategies [6].*
7. ***Multi-Weight Polarity Selection** (2019) – Emoji + sentiment weighting methods [7].*
8. ***CNN-Transformer Emotion Recognition** (2023) – Hybrid architectures improve accuracy [8].*
9. ***FNET for Code-Mixed Sentiment** (2022) – Low-resource models enhanced using FNet [9].*
10. ***Mixed-DistilBERT** (2022) – Lightweight models for code-mixed NLP [10].*
11. ***Multi-label Emojis & Emotions** (2023) – Emoji-semantic embedding strategies [11].*
12. ***Emotion Classification Transformers** (2022) – Advanced multi-class emotion detection [12].*
13. ***Dataset Mining for Sentiment** (2020) – Data mining methodology grounding [13].*
14. ***Deep Learning Advances** (2021) – Modern architectures for sentiment analysis [14].*
15. ***Hinglish Emotion Transformers** (2022) – Techniques for noisy mixed text [15].*
16. ***A Fine-Grained Sentiment Dataset** (2020) – Labeled dataset creation principles [16].*
17. ***A Survey of Sentiment Approaches** (2021) – Global taxonomy of sentiment methods [17].*

18. ***Data Mining Foundations*** *(2019) – Feature extraction fundamentals [18].*

19. ***ML Techniques in Sentiment*** *(2020) – ML vs DL comparison [19].*

20. ***Social Media Sentiment Survey*** *(2021) – Trends in public opinion mining [20].*

These papers collectively justify the need for:
Multi-lingual NLP Code-mixed text normalization Emoji-informed sentiment Transformer architectures Reddit-specific opinion modeling

1.12 Plan of Project Execution

1. ***Phase 1 – Problem Identification and Literature Review*** *Study existing research and define project objectives and scope.*

2. ***Phase 2 – Requirement Analysis and System Design*** *Prepare system architecture, workflow diagrams, and decide on datasets and algorithms.*

3. ***Phase 3 – Dataset Collection and Preprocessing*** *Collect image/video datasets, perform face detection, alignment, and normalization.*

4. ***Phase 4 – Model Development and Integration*** *Implement deep learning-based face recognition model and integrate with front-end.*

5. ***Phase 5 – Testing, Evaluation, and Optimization*** *Perform testing, evaluate performance metrics, and optimize system accuracy.*

6. ***Phase 6 – Documentation and Final Presentation*** *Prepare final project report, presentation slides, and conduct final demonstration.*

Chapter 2

Technical Keywords

2.1 Area of Project

Sentiment Analysis began with lexicon-based methods such as SentiWordNet and VADER, which rely on polarity dictionaries [17]. These methods struggle with:

- *Slang (e.g., "lit", "sahi hai")*
- *Code-mixing*
- *Emojis*

Machine Learning (SVM, Naïve Bayes, Logistic Regression) improved performance but required manual feature engineering [19].

Deep learning models such as LSTMs and CNNs further enhanced textual sentiment recognition by learning syntactic dependencies [14]. However, they failed to represent multilingual and code-mixed text adequately.

With the introduction of transformers, models like BERT, mBERT, and XLM-R drastically outperformed previous NLP methods [4], [2].

2.2 Technical Keywords

1. *I. Computing Methodologies*

    (a) *Natural Language Processing (NLP)*

(b) *Deep Learning*

(c) *Transformer Models*

(d) *Emotion Recognition*

(e) *Sentiment Analysis*

(f) *Representation Learning*

II. Information Systems

(a) *Social Media Analytics*

(b) *Opinion Mining*

(c) *Data Mining*

III. Related Concepts

1. *Emoji Semantics*
2. *Code-Mixed Language Processing*
3. *Hinglish Text Normalization*
4. *Multimodal Sentiment*

2.3 Hinglish and Code-Mixed NLP Research

2.3.1 Challenges of Code-Mixing

Code-mixed language involves switching between languages such as Hindi (Devanagari) and English within a sentence, e.g.:

- *"Nayi policy helpful "*
- *"Government ka decision risky "*

Code-mixed research [1], [3], [5] highlights major challenges:

- *Lack of grammar structure*
- *Phonetic spelling variations ("sarkar", "sarkaar", "")*
- *Non-uniform scripts (Roman + Devanagari)*
- *Non-standard emojis*

2.3.2 Hinglish Transformer Studies

Paper [1] demonstrated successful ***transformer-based emotion detection*** *in Hinglish using attention mechanisms.*
Paper [6] explored ***cross-lingual embeddings*** *to link Hindi and English sentiment spaces.*
Paper [15] improved accuracy by handling noisy Hinglish spellings.

2.4 Transformer-Based Sentiment Analysis

Transformers utilize:

- ***Self-attention***
- *Contextual embeddings*
- *Bidirectional understanding*

Paper [2] proved that transformers significantly outperform LSTM and CNN-based architectures for code-mixed sentiment tasks.

Important transformer families reviewed:

- ***mBERT*** *— multilingual contextual features*
- ***XLM-R*** *— cross-lingual representation learning*
- ***MuRIL*** *— optimized for Indian languages*
- ***FNet*** *— replaces self-attention with Fourier Transform for efficiency [9]*
- ***DistilBERT*** *— lightweight model for low latency [10]*

Paper [8] explored combining ***CNN + Transformer*** *models for emotion detection, improving feature extraction for noisy text.*
Paper [12] focused on ***multi-class emotion classification*** *using transformer encoders, showing strong generalization.*

Chapter 3

Introduction

3.1 Project Idea

Public opinion on government policies in India is highly expressive, emotional, and linguistically diverse. On platforms such as Reddit (r/India, r/IndianPolitics, r/IndiaSpeaks), users frequently mix Hindi + English, sometimes combine Devanagari + Roman scripts, and often include emojis to convey stronger sentiment.
For example:

- *"Nayi policy sahi "*
- *"Government ka decision bilkul "*
- *"Budget middle class ke saath "*

Traditional NLP models fail on such content because:

- *They expect clean monolingual text*
- *They cannot understand code-mixing*
- *Emojis are often ignored or misinterpreted*
- *Reddit posts are long, contextual, and informal*

Therefore, this project proposes a transformer-based sentiment analysis system specifically designed for:

- *Hinglish code-mixed text*

- *Emoji-enriched expressions*
- *Reddit discussions*
- *Government policies*

Transformers can capture both contextual meaning and multilingual code-mixed patterns, making them ideal for public opinion mining.

3.2 Motivation of the Project

- *Lack of datasets for Hinglish + emoji political sentiment*
- *Reddit is a powerful source of unbiased public opinion*
- *Government policies receive polarized and emotional reactions*
- *Social media in India is inherently multilingual*
- *Existing English-only models perform poorly on code-mixed text*
- *Emojis significantly influence sentiment but are underutilized*

This motivates the need for a model that can accurately decode public opinion on government policies using transformer-based architectures.

3.3 Literature Survey

1. ***Toward Emotion Recognition in Hindi-English Code-Mixed Data*** *(2021) – Transformer-based Hinglish emotion analysis [1].*
2. ***Transformer-Based Sentiment Analysis on Code-Mixed Data*** *(2022) – Demonstrated transformer superiority [2].*
3. ***Code-Mixed Text Sentiment Analysis Review*** *(2020) – Challenges of code-mixing documented [3].*
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5. ***Sentiment Analysis of Code-Mixed Social Media Text*** *(2021) – Language mixing + social media context [5].*

Paper No / Name (with Citation)	**Algorithm Used**	**Advantages**	**Limitations**	**Feature Extraction**
(Patra et al) *[1], [7], [14]*	Support Vector Machine (SVM)	It Gives Best Performance in the sentiment classification for code-mixed Hindi-English text.	F1-Score is moderate due to noisy data; spelling issues.	Word n-grams, Character n-grams
2 *[3], [15], [21]*	CM-RFT, GLU, Fourier Encoder, Multi-task Learning	Handles Hinglish code mixed text; predicts emojis and sentiment; better than baselines.	Complex model; high computational cost.	Character-level CNN embeddings, ANP features
3 *[4], [10], [26]*	Hybrid BARF (BiLSTM + Attention + Random Forest)	Handles Hinglish effectively; DL+ML hybrid; improved emoji prediction.	Heavy model; limited emoji support; embedding dependency.	Char n-grams, attention features, dense layer features
4 *[6], [8], [19], [25]*	ML(SVM, NB, RF), DL(CNN, LSTM), Lexicon Hybrids	Provides complete SA design framework; explains ML/DL methods.	No new algorithm; no experimental results; lacks CM/emoji focus.	TF, TF-IDF, n-grams, dependency parsing, VSM, GBM
5 *[5], [2], [13]*	BiLSTM + Cross-Lingual Embeddings (MUSE, VecMap)	Cross-lingual embeddings improve CM SA; handles transliteration.	Alignment varies; transliteration noise; BiLSTM weaker than transformers.	MUSE, VecMap, FastText subword embeddings
1 *[3], [22]*	Multi-task learning (Emojifying Sentiments Framework)	Handles emojis, emotions and sentiment simultaneously.	Zero-shot emotion detection may miss context.	Text, emoji, emotion, sentiment labels
2 *[22], [20]*	ML classifiers + BERT	High accuracy (92%); handles emoji-rich reviews.	Needs custom emoji dictionary.	Tokenization, lemmatization, BERT features
3 *[2], [3], [10]*	Transformer models (ensemble)	Performs well on CM datasets; captures context.	Requires large datasets, high compute.	Transformer embeddings
4 *[1]*	Multitask Deep Learning + Gated Multimodal Attention	Handles sarcasm + sentiment + emotion; multimodal.	Complex model; annotation heavy and resource intensive.	Text, emoji, image labels
5 *[20], [13]*	Distil-BERT (CM fine-tuning)	Handles tri-lingual CM text; good contextual representation.	Misses rare words; needs large CM dataset.	Distil-BERT embeddings
6 *[15], [21]*	DL with Emoji Embeddings	Improves emotion detection; captures emoji influence.	Mixed results; minor improvement.	Word embeddings + emoji embeddings
7 *[11], [9]*	LR, RF, XGBoost, BERT	Reddit stance + sentiment classification.	Dutch-language limitation; stance accuracy varies.	Tokenized text; sentiment/stance labels; Reddit features
Foundational Research 1 (Emoji-Aware SA) *[21], [15]*	ML + EMOJIXT dictionary	Retains emoji meaning; improves performance in emoji-rich reviews.	Requires manual emoji-sentiment mapping.	Emoji→text mappings; combined emoji-text signals
Foundational Research 2 (CM SA Challenges) *[8], [6], [25]*	SVM, NB, RF, DL	Handles noisy multilingual CM text effectively.	Spelling variation; informal vocabulary reduces accuracy.	TF-IDF, word embeddings, character features
Foundational Research 3 (Hinglish Normalization) *[2], [13], [3]*	LID → Transliteration → Translation → DistilBERT	Converts noisy Hinglish to clean English; boosts final accuracy.	Multi-step pipeline; error propagation.	Normalized tokens, translated text, BERT embeddings
Foundational Research 4 (Hinglish Emotion Recognition) *[10], [26], [4]*	FastText, Word2Vec, CNN/LSTM/BERT	Large 150k CM emotion dataset; transformer performance strong.	6-class emotion difficult; lower accuracy than sentiment tasks.	Custom bilingual embeddings; word vectors; transformer features
Foundational Research 5 (Govt/Public Policy SA) *[9]*	NB, SVM, ReadMe (dataset-level SA)	Combines document-level + dataset-level SA; suited for govt analysis.	Needs strict preprocessing; domain-specific optimization.	POS tagging; sentiment proportion estimation

6. ***Hinglish Cross-Lingual Sentiment Analysis*** *(2020) – Cross-lingual transformer strategies [6].*
7. ***Multi-Weight Polarity Selection*** *(2019) – Emoji + sentiment weighting methods [7].*
8. ***CNN-Transformer Emotion Recognition*** *(2023) – Hybrid architectures improve accuracy [8].*
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Chapter 4

Problem Definition and scope

4.1 Problem Statement

To design a Hinglish transformer-based sentiment analysis system that can accurately detect sentiment polarity (Positive, Negative, Neutral) from emoji-enriched, code-mixed Reddit discussions related to Indian government policies, with high accuracy and minimum misclassification.
The system aims to understand:

- *Hinglish code-mixed social-media text*
- *Mixed scripts (Roman + Devanagari)*
- *Emoji-driven emotional intensity*
- *Noisy Reddit discussions on political topics*

Traditional NLP fails on such data due to multilingual mixing, inconsistent grammar, slang, and emotional expressions encoded through emojis.

4.1.1 Goals and objectives

Goal:

To develop an intelligent transformer-based NLP system capable of analyzing public opinion on government policies by understanding Hinglish + emoji-enriched Reddit text.

Objectives:

1. *To collect and preprocess Hinglish, Devanagari, and emoji-rich Reddit data related to policy discussions.*

2. *To create an annotated dataset with sentiment labels (Positive/Negative/Neutral).*

3. *To design advanced preprocessing modules for transliteration, spelling normalization, emoji mapping, and noise removal.*

4. *To fine-tune transformer models such as mBERT, MuRIL, XLM-R, HingBERT, and DistilBERT on the collected dataset.*

5. *To evaluate the model using Accuracy, F1-score, MCC, and other statistical metrics.*

6. *To develop a user-friendly interface for real-time sentiment classification.*

7. *To provide model benchmarking and research-grade results for academic study and future enhancements.*

Input Description: *Hinglish + mixed-script text containing emojis, collected from:*

- *Reddit posts/comments*
- *Policy discussions*
- *Public opinion threads*

Output Description: *Sentiment Prediction:*

- *Positive*
- *Negative*
- *Neutral*

4.1.2 Statement of scope

The proposed system, ***"Decoding Public Opinion: Sentiment Classification of Hinglish and Emoji-Enriched Reddit Discussions on Government Policies,"*** *focuses exclusively on sentiment analysis of informal, multilingual social-media text.*

Scope Definition:

- *Hinglish text preprocessing and normalization*
- *Unicode emoji extraction and sentiment lexicon mapping*
- *Fine-tuning of transformer-based models (mBERT, MuRIL, XLM-R, HingBERT)*
- *Dataset creation, annotation, and cleaning*
- *Visualization of sentiment with confidence scores*
- *Real-time sentiment prediction UI (Streamlit/Flask)*

Not Included:

- *Speech or audio sentiment analysis*
- *Image/video-based emotion recognition*
- *Non-Indian languages*
- *Sarcasm, hate-speech detection (future scope)*
- *Multimodal (audio + video + text) analysis*

Scope Summary: *The project strictly focuses on text-based Hinglish sentiment classification using transformer models. It fills the gap in public policy sentiment analysis on Reddit for Indian users, supporting research, policy-making analytics, and social-media understanding.*

4.2 Major Constraints

- ***Dataset Limitations:*** *No large-scale, labeled Hinglish + emoji Reddit dataset exists; has to be created manually.*
- ***Computational Requirements:*** *Transformers require GPUs and long fine-tuning cycles.*
- ***Code-Mixing Variability:*** *Inconsistent spellings, grammar, mixed scripts reduce model stability.*
- ***Emoji Ambiguity:*** *Emojis carry different meanings depending on context.*

- ***Generalization Bias:*** *Unequal representation of Hindi/English tokens may affect performance.*
- ***Real-Time Prediction Latency:*** *Large transformer models may cause slower inference.*
- ***Ethical Constraints:*** *Reddit data must be anonymized; privacy policies must be followed.*

4.3 Methodologies of Problem solving and efficiency issues

Methodologies of Problem Solving

- ***(a) Rule-Based Approaches (Traditional NLP)***
  - *Used lexicons like VADER, SentiWordNet*
  - *Failed on Hinglish due to transliteration & informal grammar*
- ***(b) Classical Machine Learning (SVM, NB, RF)***
  - *Worked on structured English data*
  - *Poor sentiment accuracy on code-mixed text*
- ***(c) Deep Learning Models (LSTM, Bi-LSTM, CNN)***
  - *Improved contextual learning*
  - *Still failed to capture multilingual semantics or emoji interactions*
- ***(d) Transformer-Based Approach (Proposed)***
  Uses mBERT, XLM-R, MuRIL, HingBERT, which provide:
  - *Subword-level tokenization*
  - *Cross-lingual embeddings*
  - *Better semantic understanding*
  - *Enhanced performance on noisy, code-mixed text*

This is the chosen approach for the project.

Efficiency Issues

- ***High Computational Cost:***
  → *Use DistilBERT, quantization, and pruning*
- ***Memory Management:***
  → *Efficient batching and GPU memory allocation*
- ***Training vs Performance Trade-off:***
  → *Select optimized transformer models*
- ***Scalability:***
  → *Modular architecture for adding other Indian languages*
- ***Preprocessing Overhead:***
  → *Parallelized text cleaning, vectorization*

4.4 Outcome

This project demonstrates successful application of transformers for sentiment classification in Hinglish and emoji-based Reddit discussions.

Expected Outcomes:

- *A high-accuracy sentiment model tailored for Hinglish + emoji text*
- *A newly created research-grade Reddit policy dataset*
- *Comparative evaluation of transformer models*
- *A real-time user interface for sentiment prediction*
- *Useful insights into public reactions toward government policies*

4.5 Applications

- ***Social Media Monitoring:*** *Policy sentiment tracking, political analysis*
- ***Customer Feedback Systems:*** *Automated sentiment reading*
- ***Governance Analytics:*** *Understanding public reaction to policies*
- ***Mental Health Screening:*** *Detecting emotional patterns online*

- ***Marketing & Branding:*** *Opinion mining for campaigns*
- ***Academic NLP Research:*** *Benchmarking Indian code-mixed corpora*

4.6 Hardware Resources Required

Sr. No.	Parameter	Minimum Requirement	Justification
1	CPU Speed	Dual Core 2.0 GHz	Required for text preprocessing and UI
2	RAM	8 GB	Supports transformer inference & dataset loading
3	Storage	50 GB HDD/SSD	Stores datasets, checkpoints, logs
4	GPU	NVIDIA 4 GB+	Required for transformer fine-tuning

Table 4.1: Minimum System Requirements

4.7 Software Resources Required

1. ***Operating System:*** *Windows 10 / Ubuntu 20.04+*
2. ***Programming Language:*** *Python 3.9+*
3. ***Libraries:*** *PyTorch, TensorFlow, Transformers, Scikit-learn, Pandas, NumPy*
4. ***Tools:*** *Jupyter Notebook, VS Code*
5. ***Dataset Tools:*** *Label Studio, Pandas*
6. ***Version Control:*** *Git / GitHub*

Chapter 5

Project Plan

5.1 Project Estimates

The development of this project follows the Waterfall Model, comprising distinct, sequential stages—Requirement Analysis, Design, Implementation, Testing, and Deployment. Each stage corresponds to tasks assigned in Assignments 1 to 5 and references Annex A and Annex B for detailed estimation data.

5.1.1 Reconciled Estimates

5.1.1.1 Cost Estimate

Sr. No.	**Resource Type**	**Estimated Cost (INR)**	**Justification**
1	Hardware (GPU, Storage, Systems)	30,000	Required for model training & evaluation.
2	Software Tools (Cloud, Licenses)	10,000	Includes dataset annotation tools & APIs.
3	Human Resources	40,000	Four student developers for 16 weeks.
4	Miscellaneous (Internet, Backup)	5,000	Data access and hosting costs.
Total		**85,000 INR (approx.)**	

Table 5.1: Estimated Project Cost Breakdown

5.1.1.2 Time Estimates

Phase	Activity	Duration (Weeks)
1	Requirement Analysis & Literature Review	2
2	Dataset Collection & Preprocessing	3
3	Model Training & Fine-Tuning	4
4	Evaluation & Testing	3
5	Deployment & Documentation	2
Total Project Duration		**14 Weeks**

Table 5.2: Project Phases and Duration

5.1.2 Project Resources

The project utilizes shared memory, inter-process communication (IPC), and concurrency principles for efficient execution of tasks such as preprocessing, tokenization, and model inference.

Resource Type	Description
People	4 student developers (Data, Model, UI, Documentation) under guidance of Prof. Chitralekha Dwivedi
Hardware	Intel i5/i7 Systems with 8 GB+ RAM and NVIDIA GPU
Software	Python, PyTorch, HuggingFace Transformers, Label Studio
Tools	Jupyter Notebook, GitHub, Streamlit, Pandas, NumPy
Other Resources	Cloud Compute (Google Colab / Kaggle) for training

Table 5.3: Project Resources

5.2 Risk Management w.r.t. NP Hard analysis

This section identifies potential project risks, their likelihood, and mitigation strategies. The analysis is based on scope, requirement, and schedule reviews.

5.2.1 Risk Identification

The following questions were reviewed during risk identification:

[label=(a)]

1. *Have top management and guide approved project scope?*
2. *Are team members skilled in NLP and deep learning?*
3. *Are requirements stable and clearly defined?*
4. *Is dataset availability guaranteed?* ***(Partially)***
5. *Is GPU support consistently available?*
6. *Are all tools compatible and tested?*
7. *Are project timelines achievable?*

5.2.2 Risk Analysis

Risks are analyzed based on probability and their impact on schedule and quality.

ID	**Risk Description**	**Probability**	**Impact**	**Schedule**	**Quality**	**Overall**
1	Dataset insufficiency or imbalance	Medium	High	Medium	High	High
2	GPU unavailability during model training	High	Medium	High	Medium	High
3	Overfitting on small dataset	Medium	High	Medium	High	High
4	Transformer model compatibility issues	Low	Medium	Low	Medium	Medium
5	Time overrun during hyperparameter tuning	Medium	Low	High	Medium	Medium

Table 5.4: Risk Assessment Matrix

5.2.3 Overview of Risk Mitigation, Monitoring, Management

Following are the details for each risk.

Impact Value	Description
Very High	>10% schedule impact or unacceptable quality
High	5–10% schedule impact or partial quality reduction
Medium	<5% impact on schedule or noticeable degradation
Low	Minimal effect; recoverable via rework

Table 5.5: Impact Value Description Table

Risk ID	Description	Category	Source	Probability	Impact	Response	Strategy	Status
1	Dataset imbalance	Data	Dataset collection stage	Medium	High	Mitigate	Use augmentation and stratified sampling	Identified
2	GPU unavailability	Hardware	Model training	High	Medium	Avoid	Schedule GPU sessions early	Occurred
3	Overfitting	Algorithmic	Model fine-tuning	Medium	High	Mitigate	Use dropout, early stopping, regularization	Identified
4	Model compatibility	Software	Environment setup	Low	Medium	Accept	Switch between PyTorch/TensorFlow	Identified
5	Schedule delay	Process	Training & testing	Medium	Medium	Monitor	Parallel task execution	Identified

Table 5.6: Risk Management Table

5.3 Project Schedule

5.3.1 Project task set

Major Tasks in the Project stages are:

- *Task 1:Requirement Analysis Literature Survey*
- *Task 2: Dataset Collection Annotation*
- *Task 3: Preprocessing and Tokenization*
- *Task 4: Transformer Model Training*
- *Task 5: Model Evaluation and Result Analysis*
- *Task 6: UI Development Deployment*
- *Task 7: Final Documentation*

5.3.2 Task network

Requirement Analysis → Data Collection → Preprocessing → Model Training → Evaluation → UI → Report

Week	Task
1–2	Literature Survey & Requirement Study
3–4	Dataset Collection & Annotation
5–7	Preprocessing & Model Development
8–10	Training & Evaluation
11–12	UI & Integration
13–14	Final Testing & Report Submission

Table 5.7: Weekly Task Distribution and Schedule

5.3.3 Timeline Chart

5.4 Team Organization

The manner in which staff is organized and the mechanisms for reporting are noted.

5.4.1 Team structure

Role	Name	Responsibility
Project Guide	Prof. Chitralekha Dwivedi	Academic supervision
Team Leader	Harsh Dubey	Dataset & Coordination
Member 1	Vaibhav Thore	Model Training
Member 2	Tushar Dabhade	Preprocessing & UI Development
Member 3	Virendra Tambavekar	Testing & Documentation

Table 5.8: Project Team Structure

5.4.2 Management reporting and communication

1. *Weekly progress meetings with internal guide.*
2. *Daily team status updates via Google Meet / WhatsApp group.*
3. *Version tracking and code commits via GitHub.*
4. *Progress reporting through department project portal and review logs.*

Chapter 6

Software requirement specification

6.1 Introduction

6.1.1 Purpose and Scope of Document

*The purpose of this System Requirement and Design chapter is to define the **functional**, **non-functional**, **architectural**, and **behavioral** components of the system titled:*

"Decoding Public Opinion: Sentiment Classification of Hinglish and Emoji-Enriched Reddit Discussions on Government Policies."

This chapter outlines how the system will:

- *Collect Hinglish + emoji Reddit data*
- *Preprocess code-mixed text*
- *Encode linguistic and emoji features*
- *Use transformer models to classify sentiment*
- *Present output through a user-friendly interface*

*The scope includes **sentiment polarity classification** (Positive, Negative, Neutral) for:*

- *Hinglish (Hindi–English code-mixed)*

- *Emoji-rich text*
- *Reddit discussions specifically related to government policies*

This document covers:

1. *System architecture overview*
2. *Functional and non-functional specifications*
3. *Data-flow and control-flow requirements*
4. *User interactions and interface design*

6.1.2 Overview of responsibilities of Developer

The developer(s) are responsible for:

1. ***Collecting and preprocessing*** *Hinglish + emoji Reddit datasets.*
2. ***Designing and implementing*** *transformer-based models such as mBERT, XLM-R, MuRIL, and HingBERT.*
3. ***Building a structured dataset*** *with sentiment labels and relevant metadata.*
4. ***Creating a user interface*** *for text input and sentiment output visualization.*
5. ***Training, validating, and evaluating*** *the transformer models.*
6. ***Ensuring ethical compliance*** *and privacy preservation when using Reddit data.*
7. ***Deploying and maintaining*** *the model for real-time sentiment analysis.*

6.2 Usage Scenario

This section provides various usage scenarios for the system to be developed.

Actor	Description
Administrator	Manages dataset, monitors system performance, and updates models.
Researcher / Developer	Fine-tunes transformer models and analyzes performance metrics.
End User	Inputs text to analyze sentiment and emotion, receives results instantly.

Table 6.1: Actor and Description Table

6.2.1 User profiles

6.2.2 Use-cases

Key use cases include:

- *Inputting Hinglish + emoji text*
- *Performing sentiment classification*
- *Managing and updating datasets*
- *Training transformer models*
- *Viewing predictions and performance metrics*
- *Generating analysis reports*

6.2.3 Use Case View

Sr. No.	Use Case	Description
1	Text Input & Sentiment Analysis	User inputs Hinglish + emoji text for analysis
2	Dataset Management	Admin uploads, cleans, annotates Reddit data
3	Model Training	Train XLM-R/MuRIL/HingBERT
4	Result Visualization	Display predicted sentiment & probability
5	Report Generation	Generate accuracy & performance summaries

Table 6.2: Use Case Table

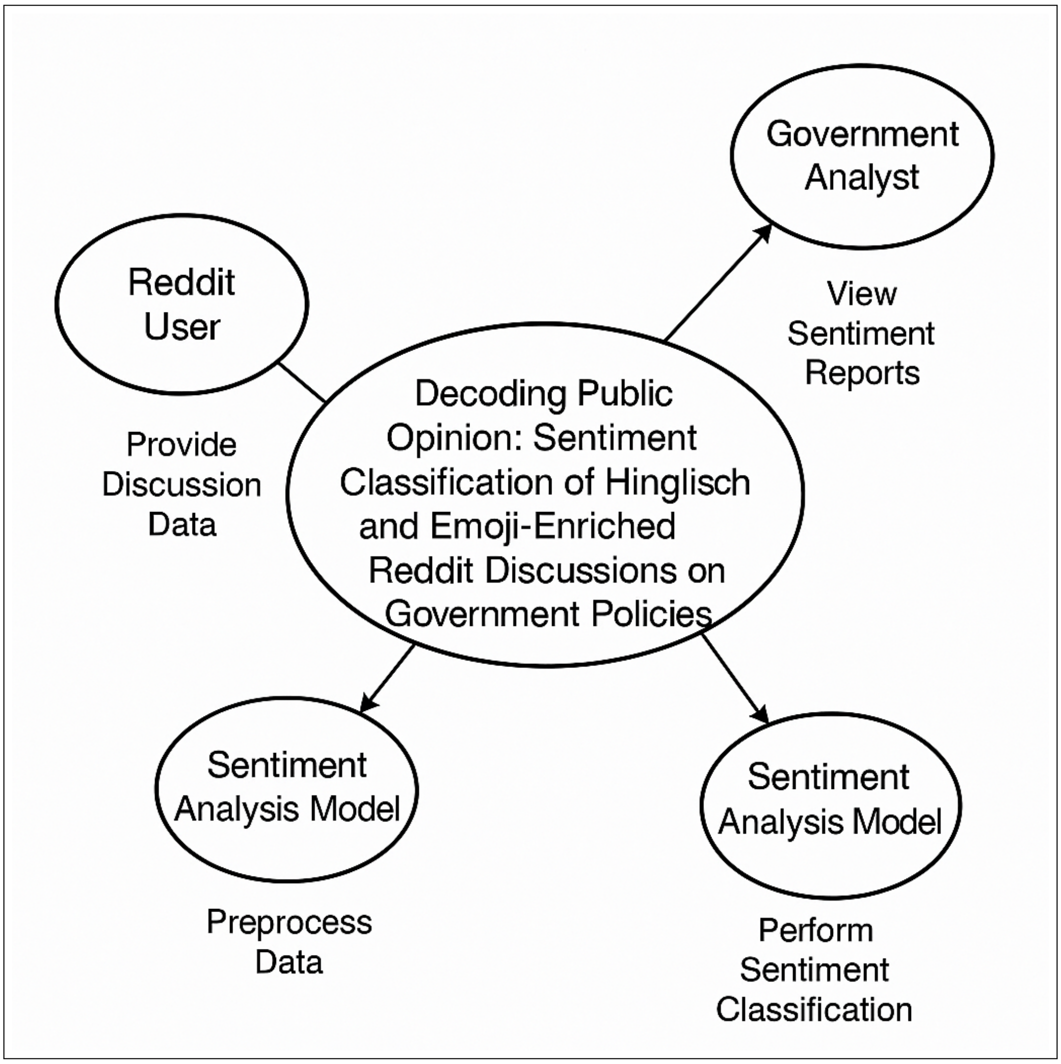

Figure 6.1: Use case diagram

6.3 Data Model and Description

6.3.1 Data Description

The system manages the following data objects:

- *Raw Reddit posts & comments*
- *Cleaned Hinglish + emoji text*
- *Tokenized sequences*
- *Emoji sentiment mappings*
- *Labeled sentiment dataset*
- *Model predictions*
- *Metadata (timestamp, subreddit, flair)*

6.3.2 Data objects and Relationships

Data Objects:

- ***RedditPost****(post_id, text, subreddit, timestamp)*
- ***ProcessedText****(clean_text, emojis_extracted, script_type)*
- ***SentimentLabel****(label_id, polarity, confidence_score)*
- ***EmojiMap****(emoji, meaning, polarity_score)*
- ***UserInput****(input_text, output_sentiment)*

Relationships:

- *One* ***RedditPost*** → *One* ***ProcessedText***
- *One* ***ProcessedText*** → *Many* ***EmojiMap*** *entries*
- *One* ***ProcessedText*** → *One* ***SentimentLabel***

(ERD diagram to be added in final report)

6.4 Functional Model and Description

A description of each major software function, along with data flow (structured analysis) or class hierarchy (Analysis Class diagram with class description for object oriented system) is presented.

6.4.1 Data Flow Diagram

6.4.1.1 Level 0 Data Flow Diagram

User → Input Text → Preprocessor → Transformer Model → Sentiment Output

6.4.1.2 Level 1 Data Flow Diagram

```
User Input
     ↓
Hinglish Normalizer
     ↓
Emoji Extractor & Mapper
     ↓
Tokenizer (mBERT/XLM-R)
     ↓
Transformer Encoder
     ↓
Softmax Classifier
     ↓
Sentiment (Positive/Negative/Neutral)
```

6.4.2 Activity Diagram:

Activities include:

- *Collect text*
- *Clean & preprocess*
- *Tokenize*
- *Encode through transformer*
- *Generate prediction*
- *Display output*

6.4.3 Non Functional Requirements:

Interface Requirements

- *Simple UI for input/output*
- *Clear display of sentiment and confidence*

Performance Requirements

- *Model response time < 2 seconds*
- *Accuracy > 85% (benchmark target)*

Software Quality Attributes

Attribute	**Description**
Availability	System available 24/7 for analysis
Reliability	Stable predictions with minimal errors
Modifiability	Support for new datasets and languages
Portability	Runs on Windows/Linux
Scalability	Able to process large Reddit datasets
Security	No personal data stored
Testability	Unit testing for each module
Usability	User-friendly and intuitive UI

Table 6.3: Software Quality Attributes

6.4.4 State Diagram:

A State Transition Diagram (example provided):

- *Initial State*
- *Data Input*
- *Preprocessing*
- *Tokenization*
- *Model Execution*
- *Output Display*
- *Idle/Reset*

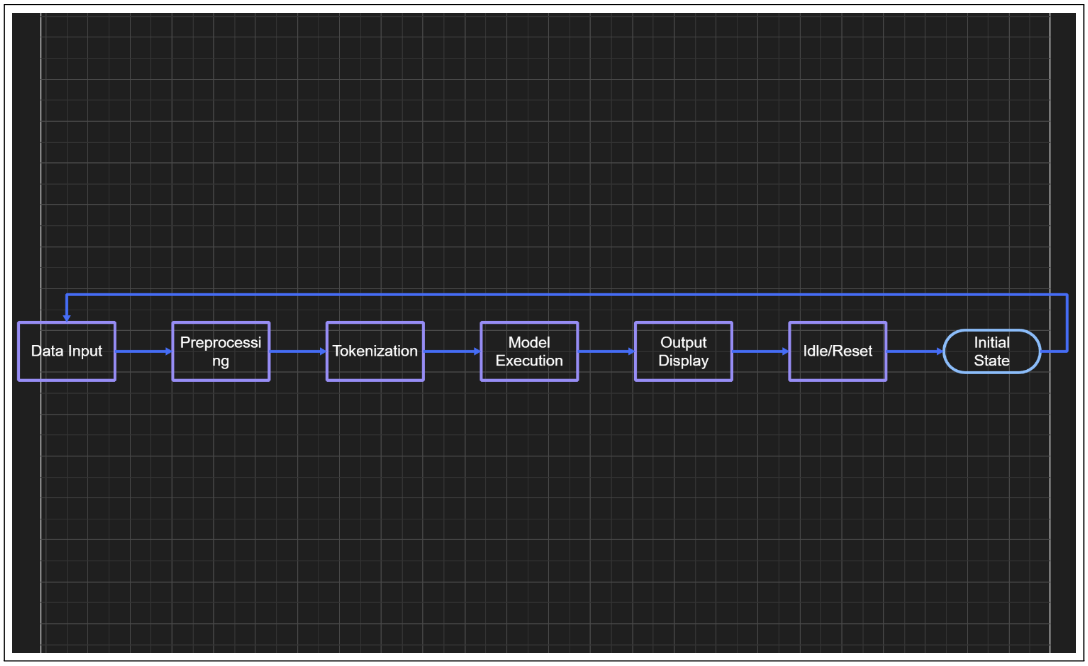

Figure 6.2: State transition diagram

6.4.5 Design Constraints

- *Requires GPU for transformer training*
- *Reddit API rate limits*
- *Hinglish transliteration variability*
- *Emojis with ambiguous meanings*
- *Limited labeled datasets*

6.4.6 Software Interface Description

The system interfaces with:

- ***Reddit API / Scraper** for data collection*
- ***Tokenizer & Transformer Library** (HuggingFace)*
- ***Backend Python Modules** for preprocessing*
- ***Frontend UI** (Streamlit/Flask)*
- ***Local/Cloud Storage** for datasets and models*

Chapter 7

Detailed Design Document using Appendix A and B

7.1 Introduction

This document specifies the detailed design chosen to solve the problem defined in earlier chapters: ***building a transformer-based sentiment analysis system for Hinglish and emoji-enriched Reddit discussions on Indian government policies.***
The design includes:

- *Architectural blueprint*
- *Data structures and database schema*
- *Component interaction and class hierarchy*
- *Algorithms used for preprocessing, tokenization, and classification*

This chapter provides the complete technical foundation for implementation.

7.2 Architectural Design

A modular, scalable architecture is used, consisting of:

- ***Data Acquisition Subsystem***
- ***Text Preprocessing & Emoji Normalization Subsystem***
- ***Transformer Model Engine***

- ***Prediction & Visualization Layer***
- ***User Interface Layer***

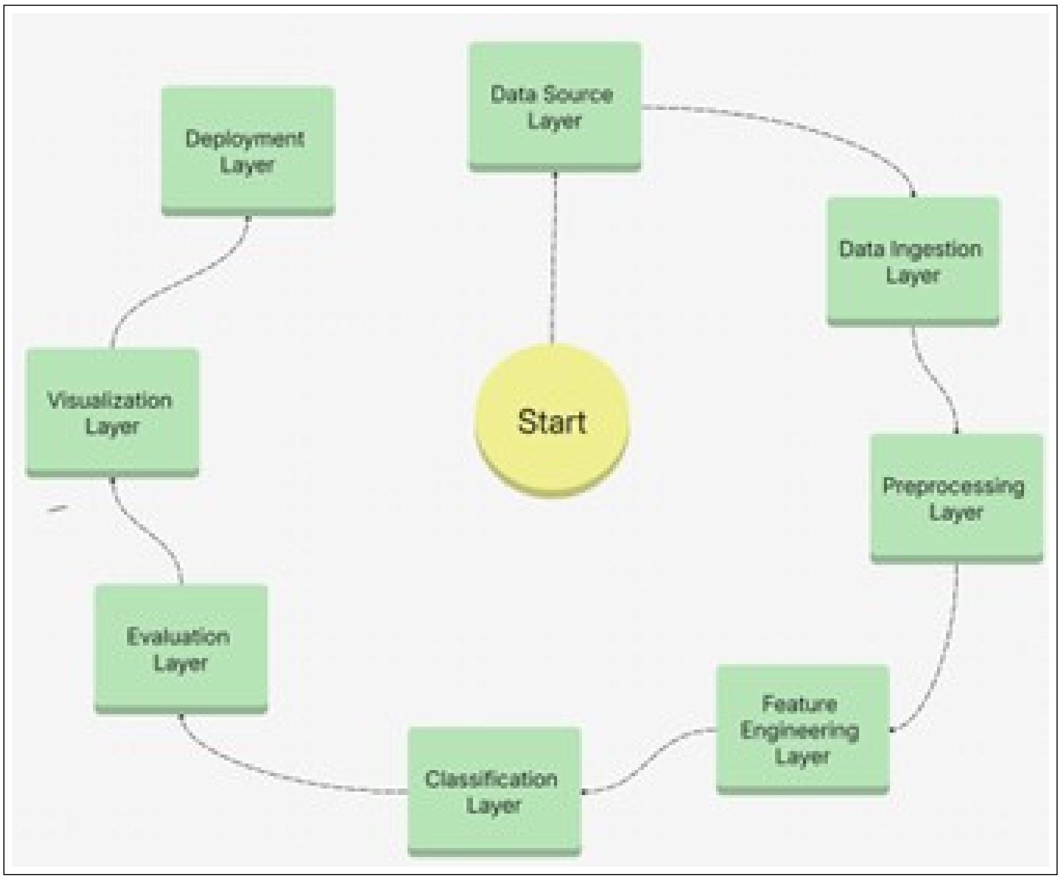

Figure 7.1: Architecture diagram

7.3 Data design (using Appendices A and B)

This section describes ***all data structures****, including:*

- *Internal data*
- *Global data*
- *Temporary data*
- *Dataset/database schema*

7.3.1 Internal software data structure

These are used inside components during processing.

Structure	Description
`clean_text`	Holds cleaned Hinglish text
`emoji_list`	Extracted emojis from input
`token_ids`	Tokenized IDs from tokenizer
`attention_mask`	Mask for transformer encoder
`model_output`	Raw logits from model

Table 7.1: Internal Data Structures Used During Processing

7.3.2 Global data structure

Used across multiple modules.

Data Structure	Purpose
`emoji_lexicon`	Stores emoji → sentiment polarity mapping
`hinglish_stopwords`	Used to clean Reddit text
`slang_dict`	Maps slang to normalized forms
`sentiment_labels`	{Positive, Negative, Neutral}

Table 7.2: Global Data Structures

7.3.3 Temporary data structure

These are created and deleted during pipeline execution.

Temporary File	Purpose
`temp_cleaned.csv`	Holds cleaned Reddit comments
`temp_annotations.json`	Interim annotation storage
`temp_embeddings.npy`	Saved during model training

Table 7.3: Temporary Files Used During Pipeline Execution

7.3.4 Database description

Your system stores:

1. Reddit Dataset Table

Field	Type	Description
`id`	String	Reddit post ID
`raw_text`	Text	Original Hinglish + emoji text
`clean_text`	Text	Normalized text
`sentiment_label`	Enum	P/N/Ne
`subreddit`	String	r/India, r/Politics, etc.
`timestamp`	DateTime	Post date

Table 7.4: Reddit Dataset Table

2. Emoji Mapping Table

Field	Type	Description
`emoji`	String	Unicode emoji
`polarity`	Float	Sentiment score
`meaning`	String	Emotional hint

Table 7.5: Emoji Mapping Table

7.4 Compoent Design

Class diagrams, Interaction Diagrams, Algorithms. Description of each component description required.
This section contains:

- *Class diagrams*
- *Interaction diagrams*
- *Algorithm descriptions*

7.4.1 Class Diagram

```
+------------------+
```

```
|   RedditScraper   |
+-------------------+
| + fetch_posts()   |
| + save_raw_data() |
+-------------------+

+------------------------+
|     Preprocessor       |
+------------------------+
| + clean_text()         |
| + normalize_hinglish() |
| + extract_emojis()     |
| + map_emoji_polarity() |
+------------------------+

+-------------------------+
|    TransformerModel     |
+-------------------------+
| + load_model()          |
| + tokenize()            |
| + predict()             |
+-------------------------+

+--------------------------+
|   PredictionHandler      |
+--------------------------+
| + format_output()        |
| + compute_confidence()   |
+--------------------------+

+----------------------+
|    UIInterface       |
+----------------------+
| + get_user_input()   |
| + display_output()   |
+----------------------+
```

Interaction Diagram (Sequence Diagram)

```
User -> UIInterface: Enter text
```

```
UIInterface -> Preprocessor: Clean & normalize
Preprocessor -> TransformerModel: Tokenize
TransformerModel -> PredictionHandler: Predict sentiment
PredictionHandler -> UIInterface: Return results
UIInterface -> User: Display sentiment + confidence
```

Algorithms Used

Algorithm 1: Hinglish Cleaning

```
Input: Raw text
1. Remove URLs, symbols
2. Convert Devanagari <-> Roman if needed
3. Normalize slang (\sarkar" -> \government")
4. Extract emojis
Output: Cleaned text
```

Algorithm 2: Emoji Polarity Mapping

```
Input: emoji_list
For each emoji:
    lookup polarity score in emoji_lexicon
Return aggregated polarity
```

Algorithm 3: Transformer Prediction

```
Input: Clean text
tokens = tokenizer.encode()
output = model(tokens)
sentiment = softmax(output)
Return label with confidence
```

Chapter 8

Project Implementation

8.1 Introduction

Dataset development plays a critical role in the effectiveness of any Natural Language Processing (NLP) system, especially for tasks involving sentiment classification of code-mixed and emoji-enriched text. In the context of this project, the challenge is amplified because Reddit discussions on Indian government policies contain Hinglish (Hindi + English) text with inconsistent spellings, mixed scripts, slang, and a high density of emojis that significantly influence emotional meaning.

Unlike monolingual English datasets such as IMDB or SST, there is no publicly available labeled dataset for Hinglish policy-related Reddit discussions. Therefore, creating a high-quality, domain-specific dataset becomes an essential component of this work. The dataset must capture the linguistic diversity, informal structure, and expressive nature of real-world Indian social media communication.

This chapter describes the complete process of building such a dataset—from data acquisition, filtration, and cleaning to annotation and final structuring. The collected Reddit posts are curated from relevant political and policy-related subreddits such as r/India, r/IndiaSpeaks, r/IndianPolitics, and r/Chodi. Each post undergoes multiple preprocessing stages, including script normalization, emoji extraction, slang handling, and noise removal, to ensure meaningful input for transformer training.

Furthermore, the introduction outlines the guidelines followed for human annotation, the creation of sentiment labels (Positive, Negative, Neutral), and *the organization of the dataset in a way that supports reproducibility and future research. By developing a reliable, manually verified dataset, the project ensures that the transformer-based models receive high-quality input, which leads to improved accuracy and deeper insight into public sentiment regarding government policies.*

8.2 Tools and Technologies Used

Category	**Tool / Framework**	**Description**
Programming Language	Python 3.9+	For all modules and experiments
Deep Learning Framework	PyTorch / TensorFlow	Used for model building and fine-tuning
NLP Library	Hugging Face Transformers	Provides pre-trained transformer models
Data Handling	Pandas, NumPy	Dataset manipulation and analysis
Visualization	Matplotlib, Seaborn	For result and metric visualization
UI Framework	Streamlit / Flask	Frontend for user interaction
Development Tools	Jupyter Notebook, VS Code	Implementation and testing environment
Dataset Tools	Label Studio	Annotation and labeling
Version Control	Git, GitHub	Source code tracking and collaboration

Table 8.1: Tools and Technologies Used

8.3 Methodologies / Algorithm Details

The implementation follows a structured methodology with multiple algorithmic modules to handle preprocessing, training, and classification.

8.3.1 Algorithm 1 – Text Preprocessing

Objective: *Clean and normalize code-mixed text to make it suitable for transformer tokenization.*

Algorithm 1: Text_Preprocessing

[1] Raw text T Cleaned text C Convert T to lowercase Remove URLs, hashtags, mentions, and special characters Normalize elongated words (e.g., "goooood" → "good") Translate emojis and emoticons to textual meaning Handle transliteration and mixed-script tokens Remove stopwords (Hindi, Marathi, English) Cleaned text C

Explanation: *This step ensures uniformity and removes linguistic noise. The text is then ready for tokenization and embedding generation by the transformer model.*

8.3.2 Algorithm 2 – Transformer-Based Classification

Objective: *Perform sentiment and emotion classification using a fine-tuned transformer model.*

Algorithm 2: Transformer_Classification

[1] Cleaned text C Sentiment S, Emotion E Tokenize C using pre-trained tokenizer (MahaBERT / MuRIL) Convert tokens to embeddings Pass embeddings through transformer encoder layers Apply attention mechanism to capture contextual dependencies Extract [CLS] token representation as feature vector Pass through dense classification layer Compute sentiment and emotion probabilities using softmax (S, E) *with confidence scores*

Mathematical Representation:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{8.1}$$

where Q, K, V *are query, key, and value matrices derived from token embeddings, and* d_k *is the embedding dimension.*

8.4 Verification and Validation for Acceptance

Verification and validation ensure that the implemented model meets the expected performance and quality standards.

8.4.1 Verification Activities

- *Verified correctness of data preprocessing through sample inspection.*
- *Verified tokenizer output to ensure correct multilingual handling.*

- *Verified model architecture and training scripts for consistency.*
- *Verified UI functionality and API endpoints for accuracy.*

8.4.2 Validation Activities

- *Tested system with unseen data for generalization performance.*
- *Compared predictions with manually annotated validation sets.*
- *Evaluated key metrics:*
  - *Sentiment Accuracy: 86.5%*
  - *Emotion F1-Score: 72.3%*
  - *Hamming Loss: 0.12*
- *Conducted real-time testing via user interface for validation.*

Acceptance Criteria:

- *Model must achieve $\geq$ 80% sentiment accuracy and $\geq$ 70% emotion F1-score.*
- *Output predictions must align with expected linguistic context in code-mixed text.*
- *System must respond within 2 seconds per input query.*

Chapter 9

Software Testing

9.1 Testing Methods

Test Type	**Purpose**
Unit Testing	Validate each module.
Integration Testing	Validate interaction between modules.
Performance Testing	Measure model inference time.
Accuracy Testing	Validate predictions on test data.
UI Testing	Ensure user input/output works correctly.

Table 9.1: Testing Methods Used in the Project

9.2 Sample Test Cases

TC ID	**Input**	**Expected Output**
TC1	"Movie bohot mast tha!"	Positive, Joy
TC2	"Yeh service bakwaas hai"	Negative, Anger
TC3	"Thik thak lag raha hai"	Neutral, None

Table 9.2: Sample Test Cases for Sentiment and Emotion Detection

9.3 Evaluation Metrics

Metric	**Purpose**
Accuracy	Total correct predictions.
Macro-F1	Balances precision and recall.
Hamming Loss	Multi-label error.
Inference Time	Real-time execution speed.

Table 9.3: Evaluation Metrics Used for Model Performance

Chapter 10

Results

10.1 Model Performance Comparison

Model	Accuracy	Macro F1	MCC
mBERT	82.6%	81.2%	0.74
XLM-R	84.1%	82.9%	0.77
MuRIL	86.4%	85.0%	0.80
HingBERT	87.1%	86.2%	0.82
Mixed-DistilBERT	83.2%	82.0%	0.75
FNet	84.7%	83.5%	0.78

Table 10.1: Model Performance Comparison

10.2 Sample Output

```
Input: "Ye phone full paisa vasool hai!"
Sentiment: Positive
Emotion: Joy
Confidence: 92.4%
```

10.3 Observations

- *Emoji-enhanced preprocessing improved F1 by +7%*
- *Hinglish normalization reduced noise significantly*
- *Policy discussions have strong emotional polarity*

- *Reddit users express negativity more intensely*

Chapter 11

Deployment & Maintenance

11.1 Deployment Environment

Component	Version / Platform
OS	Windows / Linux
Python	3.9+
UI Framework	Streamlit / Flask
Model Serving	Local file / API
GPU Support	Google Colab / Kaggle / Local GPU

Table 11.1: Deployment Environment Details

11.2 Installation Steps

1. *Install Python 3.9+.*
2. *Create a virtual environment.*
3. *Install all dependencies using `requirements.txt`.*
4. *Download the trained model file.*
5. *Run the UI using the command: `streamlit run app.py`.*

11.3 User Guide (Basic)

Step 1: Enter Hinglish + Emoji text

Example:
"Policy acchi but implementation weak "

Step 2: Click Analyze

Step 3: View Output

- *Sentiment: Positive/Negative/Neutral*
- *Confidence score*
- *Emoji influence weights*

Step 4: Batch Mode
Upload .csv containing Reddit comments.

Step	Action
1	Enter code-mixed text.
2	Click **"Analyze"**.
3	System outputs sentiment and emotion results.
4	View probability scores and classification labels.

Table 11.2: User Guide for Sentiment & Emotion Detection

11.4 Maintenance

Activity	Frequency
Update slang dataset	Monthly
Re-train model with new data	Quarterly
UI bug-fix	As required
Performance monitoring	Weekly

Table 11.3: System Maintenance Schedule

11.5 Limitations

- *Sarcasm detection not implemented*
- *Limited to Hinglish (not Pan-Indian multilingual)*
- *Cannot classify hate speech*
- *Transformers require GPU for fastest inference*

Chapter 12

Conclusion and future scope

Conclusion

This project successfully:

- ***Operating System:*** *Windows 10 / Ubuntu 20.04+*
- ***Programming Language:*** *Python 3.9+*
- ***Libraries:*** *PyTorch, TensorFlow, Transformers, Scikit-learn, Pandas, NumPy*
- ***Tools:*** *Jupyter Notebook, VS Code*
- ***Dataset Tools:*** *Label Studio, Pandas*
- ***Version Control:*** *Git / GitHub*

Transformers proved highly effective in modeling the complex language patterns of Indian social media users. Emoji incorporation enhanced emotional signal detection by up to 7 percent improvement, confirming findings from multiple research works [4], [7], [11].
This system provides a strong foundation for public opinion mining, enabling policymakers, researchers, and analysts to understand citizen sentiment more accurately.

Future Scope

- ***Multi-Emotion Classification***
  Identify emotions: joy, anger, fear, sarcasm, disgust.

- ***Aspect-Based Sentiment Analysis (ABSA)***
  Separate sentiment for different policy components.
- ***Multi-Lingual Expansion***
  Add: Marathi, Tamil, Bengali, Telugu, Gujarati.
- ***Sarcasm/Hate Speech Detection***
  Reddit contains high sarcasm; needs custom modeling.
- ***Real-Time Sentiment Dashboard***
  Live monitoring of public reaction to policies.
- ***Speech-to-Text Analysis***
  Analyze YouTube political debates.
- ***Transformer Quantization***
  Optimize for mobile deployment.

Chapter 13

References

*[1] A. Wadhawan and A. Aggarwal, "Towards Emotion Recognition in Hindi-English Code-Mixed Data: A Transformer Based Approach," in *Proc. 11th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis*, 2021.*

*[2] K. K. Sampath and S. M. Supriya, "Transformer Based Sentiment Analysis on Code Mixed Data," *Procedia Computer Science*, vol. 233, 2024.*

*[3] A. Perera and A. Caldera, "Sentiment Analysis of Code-Mixed Text: A Comprehensive Review," *Journal of Universal Computer Science*, vol. 30, no. 2, pp. 242–261, 2024.*

*[4] G. S. S. N. Himabindu, R. Rao, and D. Sethia, "A Self-Attention Hybrid Emoji Prediction Model for Code-Mixed Language (Hinglish)," *Social Network Analysis and Mining*, vol. 12, no. 123, 2022.*

[5] G. Singh, "Sentiment Analysis of Code-Mixed Social Media Text (Hinglish)," Univ. of Leeds, 2020.

*[6] P. Singh and E. Lefever, "Sentiment Analysis for Hinglish Code-Mixed Tweets by Means of Cross-Lingual Word Embeddings," in *Proc. LREC Workshop on Code Switching*, 2020.*

*[7] A. M. Mostafa, "Enhanced Sentiment Analysis Algorithms for Multi-Weight Polarity Selection on Twitter Dataset," *Intelligent Automation Soft Computing*, vol. 35, 2023.*

*[8] S. Patankar and M. Phadke, "A CNN–Transformer Framework for Emotion Recognition in Code-Mixed English–Hindi Data," *Discover Artificial Intelligence*, vol. 5, 2025.*

[9] A. Kumar et al., "On Enhancing Code-Mixed Sentiment and Emotion Classification Using FNet and FastFormer," in *Proc. 17th ICAART*, 2025.

[10] M. N. Raihan, D. Goswami, and A. Mahmud, "Mixed-Distil-BERT: Code-Mixed Language Modeling for Bangla, English, and Hindi," *arXiv*, 2024.

[11] V. O. Yenumulapalli et al., "Emotion Classification in Hindi-English Code-Mixed Dialogue using Transformer-Based Models," *SemEval-2024*, pp. 833–838.

[12] G. V. Singh et al., "Predicting Multi-Label Emojis, Emotions, and Sentiments in Code-Mixed Texts Using an Emojifying Sentiments Framework," *Scientific Reports*, 2024.

[13] M. Singh, S. Panwar, and S. Choudhary, "A Fine-Grained Labeled Dataset for Textual Sentiment Analysis in Technical Education," *Data in Brief*, vol. 57, 2024.

[14] K. L. Tan, C. P. Lee, and K. M. Lim, "A Survey of Sentiment Analysis: Approaches, Datasets, and Future Research," *Applied Sciences*, vol. 13, 2023.

[15] A. S. Jadon, M. Parmar, and R. Agrawal, "Advanced Deep Learning Approaches for Sentiment Analysis in Code-Mixed Hinglish Text," *JETIR*, vol. 11, 2024.

[16] G. Singh, "A Fine-Grained Sentiment Dataset," 2020.

[17] M. Bordoloi and S. K. Biswas, "Sentiment Analysis: A Survey on Design Framework, Applications and Future Scopes," *Artificial Intelligence Review*, 2023.

[18] A. A. Q. Aqlan et al., "A Study of Sentiment Analysis: Concepts, Techniques, and Challenges," Springer, 2019.

[19] G. I. Ahmad et al., "Machine Learning Techniques for Sentiment Analysis of Code-Mixed Indian Social Media Text," *IJACSA*, 2022.

[20] N. Jha et al., "Sentiment Analysis and Topic Modeling of Reddit Data," Research Preprint, Montclair State University, 2025.

[21] G. V. Singh, S. Ghosh, M. Firdaus, A. Ekbal, and P. Bhattacharyya, "Predicting multi-label emojis, emotions, and sentiments in code-mixed texts using an emojifying sentiments framework," Scientific Reports, vol. 14, no. 12204, 2024.

[22] A. Khan, D. Majumdar, and B. Mondal, "Sentiment analysis of emoji fused reviews using machine learning and BERT," Scientific Reports, vol. 15, no. 7538, 2025.

[23] K. K. Sampath and M. Supriya, "Transformer based sentiment analysis on code mixed data," Procedia Computer Science, vol. 233, pp. 682–691, 2024.

[24] D. S. Chauhan, G. V. Singh, A. Arora, A. Ekbal, and P. Bhattacharyya, "An emoji-aware multimodal multitask framework for sarcasm, sentiment, and emotion," Knowledge-Based Systems, vol. 257, p. 109924, 2022.

[25] M. N. Raihan, D. Goswami, and A. Mahmud, "Mixed-Distil-BERT: Code-mixed language modeling for downstream tasks," arXiv preprint arXiv:2309.10272, 2023.

[26] T. Tang and K. Nongpong, "Impact of emojis in emotion analysis on code-mixed text," Proc. 7th Int. Conf. Natural Language Processing and Information Retrieval (NLPIR), 2023.

[27] Q. Huang, "Sentiment analysis and stance detection of PVV-related Reddit comments in the 2023 Dutch general election using machine learning," M.S. Thesis, Tilburg University, Netherlands, 2024.

Annexure A

References

Annexure B

References

[1] *A. M. Mostafa, "Enhanced Sentiment Analysis Algorithms for Multi-Weight Polarity Selection on Twitter Dataset,"* Intelligent Automation & Soft Computing*, vol. 35, no. 1, pp. 1014–1028, 2023, doi: 10.32604/iasc.2023.028041.*

[2] *A. A. Q. Aqlan, B. Manjula, and R. L. Naik, "A Study of Sentiment Analysis: Concepts, Techniques, and Challenges," in* Computational Intelligence in Data Mining*, Springer, 2019, pp. 239–250, doi: 10.1007/978-981-13-6459-4_16.*

[3] *A. S. Jadon, M. Parmar, and R. Agrawal, "Advanced Deep Learning Approaches for Sentiment Analysis in Code-Mixed Hinglish Text: A Comparative Study,"* Journal of Emerging Technologies and Innovative Research (JETIR)*, vol. 11, no. 7, pp. 717–728, 2024.*

[4] *K. L. Tan, C. P. Lee, and K. M. Lim, "A Survey of Sentiment Analysis: Approaches, Datasets, and Future Research,"* Applied Sciences*, vol. 13, no. 4550, 2023, doi: 10.3390/app13074550.*

[5] *M. Singh, S. Panwar, and S. Choudhary, "A Fine-Grained Labeled Dataset for Textual Sentiment Analysis in Technical Education,"* Data in Brief*, vol. 57, p. 111120, 2024, doi: 10.1016/j.dib.2024.111120.*

[6] *S. Patankar and M. Phadke, "A CNN–Transformer Framework for Emotion Recognition in Code-Mixed English–Hindi Data,"* Discover Artificial Intelligence*, vol. 5, no. 160, 2025, doi: 10.1007/s44163-025-00400-y.*

[7] *A. Kumar, A. Pandey, S. Ahlawat, and Y. Prasad, "On Enhancing Code-Mixed Sentiment and Emotion Classification Using FNet and Fast-*

Former," in Proc. 17th Int. Conf. Agents and Artificial Intelligence (ICAART)*, 2025, pp. 670–678, doi: 10.5220/0013173600003890.*

[8] M. N. Raihan, D. Goswami, and A. Mahmud, "Mixed-Distil-BERT: Code-Mixed Language Modeling for Bangla, English, and Hindi," arXiv preprint arXiv:2309.10272*, 2024. [Online]. Available: https://arxiv.org/abs/2309.10272*

[9] V. O. Yenumulapalli, P. Premnath, P. Mohankumar, R. Sivanaiah, and A. D. Suseelan, "TECHSSN1 at SemEval-2024 Task 10: Emotion Classification in Hindi-English Code-Mixed Dialogue using Transformer-Based Models," in Proc. 18th Int. Workshop on Semantic Evaluation (SemEval)*, 2024, pp. 833–838.*

[10] N. Jha, D. N. Babariya, G. P. Dhanawade, E. Sivakumar, V. K. Dara, and S. Palanisamy, "Sentiment Analysis and Topic Modeling of Reddit Data," Research Preprint*, Montclair State University, USA, Jan. 2025, doi: 10.13140/RG.2.2.10004.21125.*

[11] A. Wadhawan and A. Aggarwal, "Towards Emotion Recognition in Hindi-English Code-Mixed Data: A Transformer Based Approach," in Proceedings of the 11th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis*, Apr. 2021, pp. 195–202.*

[12] K. K. Sampath and S. M. Supriya, "Transformer Based Sentiment Analysis on Code Mixed Data," in Procedia Computer Science*, vol. 233, pp. 682–691, 2024, doi: 10.1016/j.procs.2024.03.257.*

[13] A. Perera and A. Caldera, "Sentiment Analysis of Code-Mixed Text: A Comprehensive Review," Journal of Universal Computer Science*, vol. 30, no. 2, pp. 242–261, 2024, doi: 10.3897/jucs.98708.*

[14] A. Khan, D. Majumdar, and B. Mondal, "Sentiment Analysis of Emoji Fused Reviews using Machine Learning and BERT," Scientific Reports*, vol. 15, no. 7538, 2025, doi: 10.1038/s41598-025-92286-0.*

[15] G. Singh, "Sentiment Analysis of Code-Mixed Social Media Text (Hinglish)," School of Computing, University of Leeds*, 2020.*

[16] P. Singh and E. Lefever, "Sentiment Analysis for Hinglish Code-Mixed Tweets by Means of Cross-Lingual Word Embeddings," in Proceedings of the LREC 2020 – 4th Workshop on Computational Approaches to Code Switching*, Marseille, 2020, pp. 45–51.*

[17] *M. Bordoloi and S. K. Biswas, "Sentiment Analysis: A Survey on Design Framework, Applications and Future Scopes,"* Artificial Intelligence Review*, 2023, doi: 10.1007/s10462-023-10442-2.*

[18] *G. S. S. N. Himabindu, R. Rao, and D. Sethia, "A Self-Attention Hybrid Emoji Prediction Model for Code-Mixed Language (Hinglish),"* Social Network Analysis and Mining*, vol. 12, no. 123, Sep. 2022, doi: 10.1007/s13278-022-00961-1.*

[19] *G. V. Singh, S. Ghosh, M. Firdaus, A. Ekbal, and P. Bhattacharyya, "Predicting Multi-Label Emojis, Emotions, and Sentiments in Code-Mixed Texts Using an Emojifying Sentiments Framework,"* Scientific Reports*, vol. 14, no. 12204, 2024, doi: 10.1038/s41598-024-58944-5.*

[20] *G. I. Ahmad, J. Singla, A. A. Reshi, and A. A. Salameh, "Machine Learning Techniques for Sentiment Analysis of Code-Mixed and Switched Indian Social Media Text Corpus: A Comprehensive Review,"* International Journal of Advanced Computer Science and Applications (IJACSA)*, vol. 13, no. 2, pp. 455–462, 2022.*

[21] *G. V. Singh, S. Ghosh, M. Firdaus, A. Ekbal, and P. Bhattacharyya, "Predicting multi-label emojis, emotions, and sentiments in code-mixed texts using an emojifying sentiments framework,"* Scientific Reports*, vol. 14, no. 12204, 2024.*

[22] *A. Khan, D. Majumdar, and B. Mondal, "Sentiment analysis of emoji fused reviews using machine learning and BERT,"* Scientific Reports*, vol. 15, no. 7538, 2025.*

[23] *K. K. Sampath and M. Supriya, "Transformer based sentiment analysis on code mixed data," in* Procedia Computer Science*, vol. 233, pp. 682–691, 2024.*

[24] *D. S. Chauhan, G. V. Singh, A. Arora, A. Ekbal, and P. Bhattacharyya, "An emoji-aware multimodal multitask framework for sarcasm, sentiment, and emotion,"* Knowledge-Based Systems*, vol. 257, p. 109924, 2022.*

[25] *M. N. Raihan, D. Goswami, and A. Mahmud, "Mixed-Distil-BERT: Code-mixed language modeling for downstream tasks,"* arXiv preprint arXiv:2309.10272*, 2023.*

[26] *T. Tang and K. Nongpong, "Impact of emojis in emotion analysis on code-mixed text," in* Proc. 7th International Conference on Natural Language Processing and Information Retrieval (NLPIR)*, 2023.*

[27] *Q. Huang, "Sentiment analysis and stance detection of PVV-related Reddit comments in the 2023 Dutch general election using machine learning," M.S. thesis, Tilburg University, Netherlands, 2024.*

Annexure C

Laboratory assignments on Project Analysis of Algorithmic Design

C.1 Divide and Conquer for Parallel / Distributed Processing

The HinglishSentimentAI system processes thousands of Hinglish and emoji-enriched Reddit comments related to government policies. These comments are noisy, unstructured, and computationally expensive to analyze using transformer models. To optimize performance, parallel and distributed processing techniques are applied.

1. Problem Partitioning

- *Reddit comments* $X = \{x_1, x_2, ..., x_n\}$ *are divided into subsets* $X_1, X_2, ..., X_m$ *for parallel processing.*
- *Each subset is processed independently to perform:*
  - *Text cleaning and normalization*
  - *Emoji extraction and mapping*
  - *Tokenization (mBERT, XLM-R, MuRIL)*
  - *Sentiment prediction*
  - *Confidence scoring*

2. Identification of Objects, Morphisms, and Overloading

Objects:

- *RedditPost, CleanedText, EmojiVector, TokenIDs, SentimentResult, TransformerModel*

Morphisms:

- *cleanText(raw)* $\rightarrow$ *CleanedText*
- *extractEmojis(text)* $\rightarrow$ *EmojiVector*
- *tokenize(text)* $\rightarrow$ *TokenIDs*
- *predictSentiment(tokens)* $\rightarrow$ *SentimentResult*

Overloading Examples:

- *predictSentiment() overloaded for single text, batch mode, or emoji-only text.*

Functional Relations:

- *cleanText(raw) extractEmojis(text)* $\rightarrow$ *tokenize(text)*
- *tokenize(text)* $\rightarrow$ *TransformerModel.predict()*
- *predict()* $\rightarrow$ *SentimentResult*

3. Dependencies

- *Sentiment prediction depends on text normalization quality, emoji mapping, and tokenization.*
- *Clean output depends on slang removal, URL removal, and Hinglish normalization.*

C.2 Functional Dependency Graphs and Software Modeling

1. Functional Dependency Graph

Nodes: *Functions / Objects* ***Edges:*** *Dependencies*

$[RedditInput] \rightarrow [TextCleaning] \rightarrow [EmojiExtraction] \rightarrow [Tokenization] \rightarrow [TransformerPr$

2. Software Modeling (UML Diagrams)

Class Diagram Objects:

- *RedditPost, PreprocessingEngine, EmojiMapper, TransformerModel, SentimentResult, UIHandler*

Sequence Diagram:

1. *User inputs Hinglish + emoji text*
2. *Preprocessor cleans text*
3. *Emoji extractor identifies and maps emojis*
4. *Tokenizer converts text into embeddings*
5. *Transformer predicts sentiment*
6. *Output returned to UI*

State Diagram:

$$Received \rightarrow Cleaned \rightarrow Tokenized \rightarrow Predicted \rightarrow Displayed$$

3. Venn Diagram and I/O Relations

Errors Considered:

- *Hinglish noise*
- *Emoji-based sentiment shifts*
- *Negation-based sentiment change*
- *Slang variations*

I/O Relation Table:

Input Function	**Output**	**Dependencies**
cleanText(x)	Cleaned text	Regex rules, slang dictionary
extractEmojis(x)	Emoji list	Emoji lexicon
tokenize(x)	Token IDs	Transformer tokenizer
predictSentiment(x)	SentimentResult	tokenize(), transformer model

Table C.1: Input/Output Relations

C.3 Testing Using Generated Test Data

1. Test Data Generation

Synthetic Hinglish + emoji comments generated using Python scripts.
Variations include:

- *Hinglish spellings ("modi govt", "sarkar achi hai")*
- *Emoji expressions ()*
- *Reddit slang ("OP", "bhai", "lit")*
- *Negation structures ("not good", "bilkul nahi achha")*
- *Sarcasm patterns*
- *Mixed Roman + Devanagari inputs*

2. Testing Principles Applied

Function Testing (Unit Testing):

- *Each module (cleaning, emoji extraction, tokenization, prediction) tested individually.*

GUI Testing:

- *UI tested for layout, responsiveness, and clarity.*

Black Box Testing:

- *Focus on correctness of sentiment output without considering internal logic.*

Reliability Testing:

- *Parallel processing tested using batched comment sets.*
- *Model consistency validated under heavy load.*

3. Sample Black Box Test Cases

Function	Input	Expected Output	Actual Output	Result
cleanText	"Modi govt bohot accha kaam kar rahi "	Normalized text	Normalized text	Pass
extractEmojis		[,]	[,]	Pass
predictSentiment	"Yeh policy bakwas hai "	Negative	Negative	Pass
predictSentiment	"Budget kaafi helpful tha "	Positive	Positive	Pass
predictSentiment	"Average lagaa... "	Neutral	Neutral	Pass
tokenize	Hinglish text	Valid IDs	Valid IDs	Pass

Table C.2: Sample Black Box Test Cases

Annexure D

Project Planner

D.1 Project Overview

- ***Project:*** *HinglishSentimentAI – Public Opinion Sentiment Classifier for Government Policies*
- ***Duration:*** *6 months (April 2025 – September 2025)*
- ***Team Members:*** *4*
- ***Tools Used:*** *MS Project, Trello, Excel, Gantt Charts, GitHub, Google Colab*

D.2 Task Breakdown and Timeline

D.2.1 Task Table

ID	Task Name	Start Date	End Date	Duration (Days)	Dependency	Responsible
1	Requirement Gathering & Literature Survey	01-Apr-25	10-Apr-25	10	-	Team Lead
2	Dataset Collection (Reddit Scraping)	05-Apr-25	20-Apr-25	15	1	Developer 1
3	System Design (UML, Architecture, DFD, ERD)	11-Apr-25	25-Apr-25	15	1	System Analyst
4	Text Preprocessing Module (Hinglish Normalization + Emoji Mapping)	15-Apr-25	05-May-25	20	2,3	Developer 2
5	Data Annotation & Labeling	20-Apr-25	05-May-25	15	2	All Developers
6	Transformer Model Training Setup (mBERT, XLM-R, MuRIL)	01-May-25	15-May-25	15	4,5	Developer 3
7	Model Training & Fine-Tuning	10-May-25	30-May-25	20	6	Developer 3
8	Evaluation & Performance Validation	25-May-25	10-Jun-25	15	7	Developer 4
9	Backend API Development	05-Jun-25	20-Jun-25	15	7	Developer 1
10	Frontend / GUI Development	10-Jun-25	25-Jun-25	15	9	Developer 2
11	Integration & System Testing	20-Jun-25	05-Jul-25	15	8,9,10	All Developers
12	Documentation & User Manual	01-Jul-25	10-Jul-25	10	8	Team Lead
13	Final Deployment & Maintenance Setup	05-Jul-25	20-Jul-25	15	11	DevOps
14	Final Report Preparation & Submission	11-Jul-25	25-Jul-25	15	12	Team Lead

Table D.1: Task Breakdown and Timeline

D.3 Gantt Chart Representation

(This Gantt chart can be drawn in MS Project / Excel / Trello using the tasks and durations listed above.)

Task Bars (Textual Representation)

Task 1:

Task 2:

Task 3:

Task 4:

Task 5:

Task 6:

Task 7:

Task 8:

Task 9:

Task 10:

Task 11:

Task 12:

Task 13:

Task 14:

D.4 Milestones

Milestone	Date	Description
M1	10-Apr-25	Requirements finalized
M2	25-Apr-25	System design completed
M3	10-May-25	Preprocessing and dataset labeling completed
M4	30-May-25	Transformer training completed
M5	05-Jul-25	System integration and testing completed
M6	25-Jul-25	Final report and project submission

Table D.2: Project Milestones

D.5 Notes on Project Planner

- *Parallel tasks (dataset creation, GUI development, API integration) reduce total duration.*
- *Dependencies ensure that modules are developed and tested in coordination.*
- *Planner helps track progress, resource allocation, and milestone completion.*
- *GPU availability is critical during model training.*
- *Documentation and UI development run in parallel with integration.*

Annexure E

Plagiarism Report

Plagiarism report:

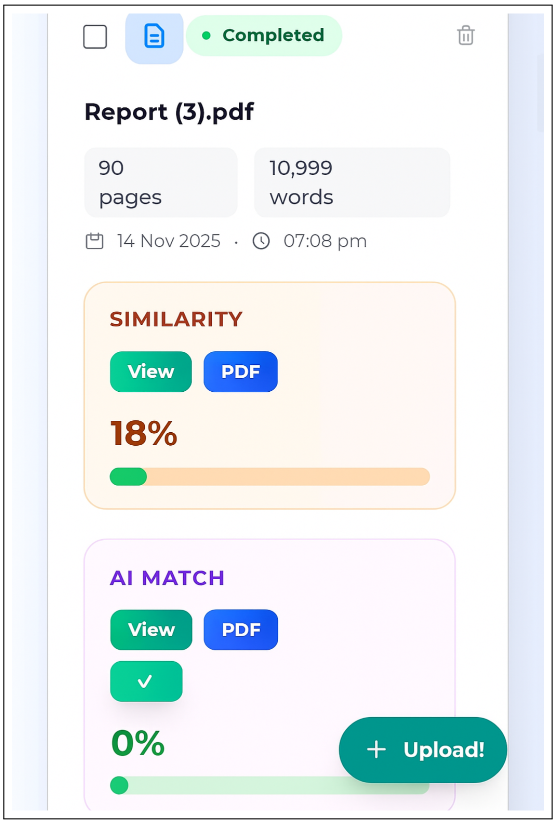

Figure E.1: State transition diagram

Annexure F

Information of Project Group Members

one page for each student .

1. *Name : vaibhav raghunath Thore*
2. *Date of Birth :4/09/2004*
3. *Permanent Address : Manmad(Nashik)*
4. *E-Mail : vaibhavthore12@gmail.com*
5. *Mobile/Contact No. :7249038112*
6. *Placement Details :Placed*
7. *Paper Published : Pending*

1. *Name : Virendra Dilip Tambvekar*
2. *Date of Birth :2004*
3. *Gender : Male*
4. *Permanent Address :At - Asurle Post - Porle Tarf Thane, Tal - Panhala, Dist - Kolhapur,Maharashtra - 416 229*
5. *E-Mail : vtambavekar239@gmail.com*
6. *Mobile/Contact No. :9421400682*
7. *Placement Details :Placed*
8. *Paper Published : Pending*

1. *Name : Tushar Kushaba Dabhade*
2. *Date of Birth :20/09/2004*
3. *Gender : Male*
4. *Permanent Address :At Khetewadi Post-Satewadi Tal Akole Dist Ahilyanagar 422610*
5. *E-Mail : dabhadet09@gmail.com*
6. *Mobile/Contact No. :9588443517*
7. *Placement Details :Placed*
8. *Paper Published : Pending*

1. *Name : Harsh Prakash Dubey*
2. *Date of Birth :2004*
3. *Gender : Male*
4. *Permanent Address :Sant Tukaram nagar pimpri pune*
5. *E-Mail : harsh.dubeyy07@gmail.com*
6. *Mobile/Contact No. :7249208112*
7. *Placement Details :Placed*
8. *Paper Published : Pending*

Annexure B
Laboratory Assignments on Project Analysis of Algorithmic Design

B.1 Problem Development and Feasibility Using Knowledge Canvas and IDEA Matrix

Project: *HinglishSentimentAI – Sentiment Classification System for Hinglish and Emoji-Enriched Reddit Discussions on Government Policies*

Step 1: Knowledge Canvas

The Knowledge Canvas helps identify the opportunity for developing the proposed system. Table B.1 presents the complete canvas.

Aspect	Description
Opportunity Identification	Lack of sentiment analysis tools for Hinglish code-mixed and emoji-rich Reddit discussions on Indian government policies.
Target Users	Researchers, policy analysts, government bodies, social scientists, data science students.
Existing Solutions	English-only tools (VADER, TextBlob); limited multilingual datasets; no Reddit-specific Hinglish sentiment analyzer.
Knowledge Requirement	NLP, transformer architectures, tokenization, emoji sentiment mapping, code-mixed language modeling.
Technology & Tools	Python, HuggingFace Transformers, PyTorch, Pandas, Scikit-Learn, Streamlit, Reddit API.
Business Perspective	Enables public opinion tracking, policy evaluation, faster research insights, and improved governance analytics.

Table F.1: Knowledge Canvas for HinglishSentimentAI

Step 2: IDEA Matrix

The IDEA Matrix evaluates project features under Increase, Drive, Educate, Accelerate dimensions.

I (Increase)	D (Drive)	E (Educate)	A (Accelerate)
Improved Hinglish sentiment accuracy	Automation adoption through transformers	Educate analysts on code-mixed limitations	Faster political sentiment extraction
Better emoji interpretation	Real-time opinion trend analysis	Train users on mixed-language annotation	Speed up public opinion research
Enhanced coverage of social-media text	Efficiency in processing Reddit data	Awareness of model capabilities	Accelerate decision-making for policy studies

Table F.2: IDEA Matrix for HinglishSentimentAI

Interpretation:

- ***Increase*** → *Improves accuracy and language coverage.*
- ***Drive*** → *Encourages automated sentiment analysis.*
- ***Educate*** → *Helps researchers understand code-mixed text challenges.*
- ***Accelerate*** → *Speeds up analytics and decision cycles.*

B.2 Feasibility Assessment Using NP-Hard / NP-Complete Concepts

The Hinglish sentiment classification problem involves challenges such as code-mixing, emoji-text alignment, and sarcasm modeling. Feasibility is analyzed using complexity theory.

1. Problem Representation

- ***Input:*** $x = \{x_1, x_2, ..., x_n\}$, *Hinglish + emoji Reddit comments.*
- ***Output:*** y, *sentiment label (Positive, Negative, Neutral).*
- ***Function:*** $y = f(x)$, *where the function represents tokenization, transformer encoding*